Label-Free Recovery of a Latent Routing Partition from m-Response Profiles

Formalization, Instantiations, and Identifiability Theory

Working note

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Abstract

We formalize the problem of recovering, label-free and from black-box query access alone, the latent routing partition of a model whose behavior is gated into K hidden modes. The auditor observes only m-response profiles—the model's responses to small perturbations of each input. We show that the profile is an *induced separable admixture* whose mixing weights are a geometry-coupled functional of the gate, and whose anchor (unperturbed) coordinate is a pure draw from the input's mode. We give two instantiations, a static adversarial scaffold and a benign mixture-of-experts, unified by a formal gate–manifold alignment axis. We then state the identifiability program: a recovery theorem routed through near-separable nonnegative matrix factorization (T2), an estimator guarantee (T2′), a probe-conditioning phase boundary (T3), and a graceful-degradation result under anchor impurity (T4). The adaptive adversary is deliberately out of scope and is the subject of follow-up work; the bridge to it is the anchor-purity parameter of T4.

1 Setup and the recovery problem

Model under audit. Let $f : \mathcal{X} \to \mathcal{Y}$ be a function accessible only by query (no internals, gradients, or weights). We posit that f is internally organized by a *latent gate*

$$g : \mathcal{X} \to [K], \qquad [K] = \{1, \dots, K\},$$

assigning each input to one of K *behavioral modes*, together with per-mode *response laws* $\{R_k(\,\cdot\,; x)\}_{k\in[K]}$, so that a single query behaves as $f(x) \sim R_{g(x)}(\,\cdot\,; x)$. The gate g is a property of the deployed model, never of the auditor, and is never observed.

Point-mass responses (primary model). Throughout we take each mode's response to be *deterministic given the input*, up to observation noise:

$$R_k(\,\cdot\,; x) \;=\; \delta_{\,f_k(x)} \;\; (\text{plus noise } \eta,\ \mathbb{E}\eta = 0,\ \mathbb{E}\,\|\eta\|^2 = \tau_{\text{obs}}^2), \tag{1}$$

for mode functions $f_1, \dots, f_K : \mathcal{X} \to \mathcal{Y} \subseteq \mathbb{R}^d$. This matches the scaffold and MoE constructions below, where each branch/expert is a fixed function. (The fully distributional case is a generalization; see Remark 7.)

Probe and response profile. The auditor chooses a *probe policy* Q that, given an anchor input x, draws m perturbations $\delta_1, \dots, \delta_m \sim Q(\cdot \mid x)$. We write $Q = Q_\sigma$ when the probe has a scale parameter σ (e.g. $\delta \sim \mathcal{N}(0, \sigma^2 I)$). Querying f at the anchor and the perturbed points yields the *m-response profile*

$$\Phi_i \;=\; \big(\underbrace{f(x_i)}_{\text{anchor}},\; f(x_i+\delta_{i1}), \dots, f(x_i+\delta_{im})\big) \;\in\; \mathcal{Y}^{m+1}. \tag{2}$$

The estimand. Given only $\{\Phi_i\}_{i=1}^n$ for a queried sample $x_1, \dots, x_n$, recover the *routing partition* $z_i := g(x_i) \in [K]$. As developed below, the natural *primary* estimand is the soft composition $\pi_i \in \Delta^{K-1}$; the hard partition and a per-instance anomaly flag are read-outs of it (Section 5).

2 The induced admixture (T1)

The central modeling fact is that *each perturbed query is routed by its own location*, not by the anchor's mode.

Definition 1 (Local routing composition)**.** The local routing composition at x_i under probe Q is

$$\pi_i(k) \;:=\; \mathbb{P}_{\delta \sim Q(\cdot \mid x_i)}\big[g(x_i+\delta) = k\big], \qquad k \in [K], \quad \pi_i \in \Delta^{K-1}. \tag{3}$$

The j-th perturbed coordinate then has component label $a_{ij} = g(x_i + \delta_{ij}) \sim \text{Categorical}(\pi_i)$, and $f(x_i + \delta_{ij}) \sim R_{a_{ij}}(\,\cdot\,; x_i + \delta_{ij})$.

Definition 2 (Induced separable admixture)**.** Under (1)–(3) the profile is generated by

$$\Phi_i \;\sim\; \underbrace{R_{z_i}(\,\cdot\,; x_i)}_{\textit{pure anchor}} \;\otimes\; \bigotimes_{j=1}^{m} \Big[\textstyle\sum_{k\in[K]} \pi_i(k)\, R_k(\,\cdot\,; x_i + \delta_{ij})\Big]. \tag{4}$$

The anchor coordinate is a single draw from R_{z_i}, since $\delta = 0$ cannot change the gate; it is *pure*. The remaining m coordinates form an *admixture* with mixing weights π_i.

Remark 1 (Why this is not a simple mixture). A simple mixture would assert $\Phi_i \mid z_i \sim P_{z_i}$ for a fixed per-mode profile law. That holds only if π_i is a deterministic function of z_i (no perturbation crosses the gate). In general π_i is a *geometry-coupled* functional of x_i through (3), varying continuously with the position of x_i relative to the gate boundary. The admixture structure is *forced by the query mechanism*, not assumed; this is the key structural difference from latent-label-shift / topic-model recovery, where the mixing proportions are exogenous and i.i.d.

Matrix form. Fix a feature map $\psi : \mathcal{Y} \to \mathbb{R}^V$ and summarize each mode's local response by $w_k(x) := \mathbb{E}_\eta[\psi(f_k(x)+\eta)]$. For a localized analysis around a region where the w_k are approximately constant, write $w_k := \mathbb{E}_x[w_k(x) \mid \text{region}]$, collect the *mode matrix* $W = [w_1, \dots, w_K] \in \mathbb{R}^{V\times K}$, and the *composition matrix* $H = [\pi_1, \dots, \pi_n] \in \mathbb{R}^{K\times n}$ (columns in Δ^{K-1}). Summarizing each profile by its empirical coordinate-mean $M_{\cdot i} := \frac{1}{m}\sum_j \psi\big(f(x_i+\delta_{ij})\big)$ gives

$$M \;=\; WH \;+\; N, \qquad M \in \mathbb{R}^{V\times n}, \tag{5}$$

where N is a mean-zero error with $\mathbb{E}\,\|N_{\cdot i}\|^2 = O\big((\tau_{\text{obs}}^2 + \sigma_w^2)/m\big)$, σ_w^2 the within-mode response variance over the probe neighborhood. Recovery of the routing partition is recovery of H; the anchor supplies a per-column pure observation, discussed in Section 6.

3 Two instantiations

3.1 Static adversarial gate: feature-thresholded redlining ($K = 2$)

Example 1 (Feature-threshold scaffold)**.** Let $b : \mathcal{X} \to [0, 1]$ be a protected feature (neighborhood minority share) and τ a threshold. The deployed model is

$$f(x) \;=\; h(x) \;-\; 0.3 \cdot \mathbf{1}\big[\, b(x) \geq \tau \,\big], \qquad g(x) = \mathbf{1}[\, b(x) \geq \tau \,] \in \{1, 2\},$$

with $f_1 = h$ (full valuation) and $f_2 = h - 0.3$ (penalized log-price). **Anchor purity is automatic** for any deterministic location gate: the unperturbed input is routed by its own position. Real inputs populate both sides of $b = \tau$, so the partition over real data is non-trivial: recovering z_i is recovering the redlining map. The composition is $\pi_i(2) = \mathbb{P}_\delta[\, b(x_i + \delta) \geq \tau \,]$. Structurally this gate is identical to the benign competence switch of Example 3; adversariality is intent and concealment, not geometry.

Remark 2 (Relation to the Slack OOD scaffold). Slack's original construction gates on *probe detection*: an OOD detector $d(x)$ sends "real" inputs to a biased model and suspected synthetic queries to an innocuous one. Under that gate every real instance is mode-biased, so the per-instance partition over real data is trivial and the recovery target shifts to query routing. The full fooling-LIME/SHAP construction stacks both gates, yielding $K = 3$ effective modes: (real, $b < \tau$) $\to h$; (real, $b \geq \tau$) $\to h - 0.3$; (flagged probe) $\to$ innocuous. The formalism covers the stack directly; the two boundaries are crossed by different probe components (Section 4).

3.2 Latent benign gate: mixture-of-experts (K experts)

Example 2 (MoE router)**.** Let $g(x) = \arg\max_{k \in [K]} r_k(x)$ select expert z_i, with expert functions f_k. The anchor is a pure draw from the expert owning x_i. The composition is $\pi_i(k) = \mathbb{P}_\delta[\arg\max_\ell r_\ell(x_i + \delta) = k]$. **Asymmetry vs. the scaffold:** deep in expert z_i's region every perturbation stays in expert z_i, so $\pi_i = e_{z_i}$ is a simplex vertex and the admixture degenerates—the anchor alone identifies z_i, but there is no mixing to exploit. The admixture is nondegenerate *only near router boundaries*; hence benign routing recovery is dense at boundaries and obtained in the interior by propagation (Section 6). Here K is the number of experts and is generally unknown (Section 9).

Example 3 (Benign competence/quantization switch)**.** A served model may route confident inputs to a fast head and uncertain inputs to a large head: $g(x) = \mathbf{1}[c(x) < \theta]$ for a confidence score c. The boundary lies along a level set of c, typically *tangent* to the data manifold—a competence-aligned gate in the sense of Section 4.

4 The gate–geometry axis (T3, structural part)

We make "competence-aligned vs. manifold-aligned" a formal scalar. Let $\mathcal{M} \subset \mathcal{X}$ be the data manifold with tangent space $T_x\mathcal{M}$, and suppose near a boundary point x the gate boundary is smooth with unit normal $\nu(x)$.

Definition 3 (Gate–manifold alignment)**.** The alignment at a boundary point x is $\alpha(x) = \big\|\mathrm{proj}_{T_x\mathcal{M}}\, \nu(x)\big\| \in [0, 1]$. $\alpha \approx 0$: the boundary normal is orthogonal to the manifold (the boundary runs *along* the manifold, so on-manifold probes rarely cross; competence/MoE-like). $\alpha \approx 1$: the normal lies in the manifold (the boundary cuts *across* it, so probes cross easily; scaffold-like).

Remark 3. α controls how much *on-manifold* probe mass crosses the boundary at fixed scale σ, hence the off-vertex magnitude of π_i for boundary-adjacent i. The feature-threshold scaffold of Example 1 sits near $\alpha \approx 1$ (its boundary, a level set of a real feature, cuts across the manifold); the benign MoE near small α. A third type, the OOD-detector gate of Remark 2, *wraps around* the manifold: its normal points off-manifold ($\alpha \approx 0$), so on-manifold probes never cross it, while ambient Gaussian probes cross easily through the many off-manifold directions. The general quantity is therefore the *probe-relative crossing scale* $\sigma_{\text{eff}}(x) = \sqrt{\nu(x)^\top \Sigma\, \nu(x)}$ for probe covariance Σ, with crossing probability $\approx \Phi(-s/\sigma_{\text{eff}})$ at signed distance s; manifold alignment is the on-manifold special case $\sigma_{\text{eff}} = \sigma\alpha$. Probe *support*, not only scale, is thus an identifiability knob. Section 7 ties σ_{eff} to the conditioning of H and thereby to recoverability, making the single-paper, two-instantiation scope a statement about one axis rather than two case studies.

5 What we output

The estimator returns, per instance, a soft composition $\widehat{\pi}_i \in \Delta^{K-1}$. From it:

- **Regime classification:** $\widehat{z}_i = \arg\max_k \widehat{\pi}_i(k)$ (the hard partition; recovered up to a global label permutation).
- **Anomaly flag** (binary collapse): for a designated adverse mode $k^\star$, flag i iff $\widehat{\pi}_i(k^\star) \geq t$. In the scaffold, $k^\star$ is the biased mode and the flag marks instances the model treats adversarially.
- **Soft routing map:** report $\widehat{\pi}_i$ itself, which encodes boundary proximity (interior points are near a vertex; boundary points are mixed).

Guarantees below are stated for $\widehat{\pi}_i$; the other two outputs inherit them by post-processing (Corollary 1).

6 Identifiability and recovery (T2, T2′)

We route recovery through *near-separable NMF*, whose guarantees are deterministic in the mixing weights—this is what kills the "proportions must be i.i.d." objection inherited from Dirichlet-based topic models.

Definition 4 (Separability / pure instances)**.** Factorization (5) is *separable* if for each mode $k \in [K]$ there exists an instance i with $\pi_i = e_k$ (a column of H equal to the k-th vertex). Such an i is a *pure instance* for mode k; its profile column $M_{\cdot i} \approx w_k$.

Remark 4 (The anchor relaxes separability). We need not hope a fully pure *profile* exists: the anchor coordinate of *every* instance is a pure draw from its own mode (Def. 2). Thus each instance contributes one guaranteed-pure observation $\psi(f(x_i))$ of w_{z_i}. Separability of W's columns is therefore witnessed by the anchor set $\{\psi(f(x_i))\}_i$, which contains pure samples of every mode that appears as some z_i. This is strictly weaker than the topic-model anchor-*word* assumption (a coordinate pure across the corpus); we instead have a pure *coordinate per instance*.

Assumption 1 (Recovery conditions)**.** (i) **Separated modes:** W has full column rank K with $\sigma_{\min}(W) > 0$. (ii) **Composition diversity:** the columns of H are not contained in any proper face's cone beyond what separability allows; equivalently the proportion second-moment $G := \frac{1}{n} H H^\top \in \mathbb{R}^{K \times K}$ is nonsingular, with condition number $\kappa(H) := \sigma_{\max}(H)/\sigma_{\min}(H)$. (iii) **Anchor purity:** a $(1-\varepsilon)$ fraction of anchors are pure (Section 8 treats $\varepsilon > 0$; here $\varepsilon = 0$).

Theorem 1 (Exact identifiability, noiseless)**.** *Under Assumption 1 with $\varepsilon = 0$ and $N = 0$, the routing composition H and mode matrix W in (5) are identifiable up to a permutation of $[K]$. Concretely, the vertices $\{w_k\}$ are exactly the extreme rays of* cone(M) *recovered by the anchor set, and given W, each π_i is the unique solution of $M_{\cdot i} = W\pi_i$, $\pi_i \in \Delta^{K-1}$.*

Proof. By (5) with $N = 0$, every column $M_{\cdot i} = W\pi_i$ lies in $\mathrm{conv}\{w_1, \ldots, w_K\}$. By separability (Def. 4), for each k some column equals w_k, so the K vertices of the simplex $\mathrm{conv}\{w_k\}$ are present among the data columns; by Remark 4 the anchor observations exhibit them explicitly. Full column rank of W (Assumption 1(i)) makes the vertex set a basis, so the convex hull is a nondegenerate $(K-1)$-simplex whose vertices are its unique extreme points; these are determined by the data up to relabeling. Given W, full column rank yields a unique π_i with $M_{\cdot i} = W\pi_i$ and $\sum_k \pi_i(k) = 1$. Both W and H are thus pinned up to the simultaneous permutation $w_k \mapsto w_{\rho(k)}$, $\pi_i(k) \mapsto \pi_i(\rho(k))$. □

Theorem 2 (Robust estimator guarantee, noisy)**.** *Let $\widehat{W}$ be produced by the Successive Projection Algorithm (SPA) on the columns of $M = WH + N$, and $\widehat{\pi}_i$ by simplex-constrained least squares against $\widehat{W}$. Suppose Assumption 1 ($\varepsilon = 0$) holds and the per-column noise satisfies*

$$\max_i \|N_{\cdot i}\| \;\le\; \epsilon \;\le\; \mathcal{O}\!\left(\frac{\sigma_{\min}(W)}{\sqrt{K}\,\kappa^2(W)}\right).$$

Then SPA identifies all K mode directions up to error $\mathcal{O}(\epsilon\,\kappa^2(W))$, and consequently

$$\max_i \|\widehat{\pi}_i - \pi_i\| \;\le\; \mathcal{O}\!\left(\frac{\epsilon\,\kappa^2(W)}{\sigma_{\min}(W)}\right).$$

Since $\epsilon = \mathcal{O}\big(\sqrt{(\tau_{\mathrm{obs}}^2 + \sigma_w^2)/m}\big)$, the bound vanishes as $m \to \infty$.

Proof sketch. The first claim is the Gillis–Vavasis robustness theorem for SPA on near-separable matrices: with W full column rank and bounded column noise, the greedy max-norm selection returns columns within $\mathcal{O}(\epsilon\,\kappa^2(W))$ of the true vertices. Crucially this statement places *no distributional assumption on H*; only W's conditioning enters, so geometry-coupled (deterministic, correlated) compositions are admissible. Propagating the vertex error through the simplex-constrained least-squares solve, whose conditioning is governed by $\sigma_{\min}(W)$, gives the stated $\widehat{\pi}_i$ bound. The noise scale ϵ is the Monte-Carlo error of the m-coordinate average in (5), of order $\sqrt{(\tau_{\mathrm{obs}}^2 + \sigma_w^2)/m}$. □

Corollary 1 (Read-out guarantees)**.** *If $\max_i \|\widehat{\pi}_i - \pi_i\| \le \gamma$ then: (a) every interior instance with $\pi_i(z_i) \ge \frac{1}{2} + \gamma$ is classified correctly, $\widehat{z}_i = z_i$ (up to permutation); (b) the anomaly flag at threshold t has no false negatives among instances with $\pi_i(k^\star) \ge t + \gamma$ and no false positives among those with $\pi_i(k^\star) \le t - \gamma$.*

Proof. Immediate from $\|\widehat{\pi}_i - \pi_i\|_\infty \le \gamma$ and the definitions of arg max and threshold read-outs. □

Remark 5 (This is the tractability hinge). Theorem 1 says the problem is *well-posed* (the partition is a function of the observable law); Theorem 2 says a concrete polynomial-time estimator *attains* it under explicit conditioning. The binding empirical question is whether the conditions of Assumption 1(ii) hold for realistic gates—this is exactly what the probe controls, and is the content of T3.

7 Probe design changes identifiability (T3, quantitative)

T3 states that probe design enters the recovery bound through the conditioning $\kappa(H)$, and that this is governed by the probe scale σ and the gate–manifold alignment α of Definition 3—*not* merely by signal-to-noise.

Proposition 1 (Conditioning is probe-controlled)**.** *Fix a gate with boundary alignment α near a boundary region, and a Gaussian probe Q_σ. There is a scale band $[\sigma_{\min}(\alpha), \sigma_{\max}(\alpha)]$ such that $\sigma_{\min}(G)$, the smallest eigenvalue of the proportion second moment $G = \frac{1}{n} HH^\top$, is bounded below by a positive constant for σ in the band and tends to 0 outside it:*

- *$\sigma \to 0$: every $\pi_i \to e_{z_i}$ (no crossings), so H's columns collapse onto vertices; G approaches a diagonal occupancy matrix with one underrepresented mode if a region is interior-dominated—rank may drop, recovery via mixing vanishes (only anchors inform).*
- *$\sigma \to \infty$: every $\pi_i \to \bar{\pi}$ (probe spans all modes equally), so H's columns collapse onto a common centroid; $\sigma_{\min}(G) \to 0$ and $\kappa(H) \to \infty$—recovery impossible.*
- *intermediate σ: columns of H populate the interior of the simplex with diversity, $\sigma_{\min}(G)$ bounded away from 0.*

The band widens as $\alpha \to 1$ (transverse boundary: crossings occur at smaller σ) and narrows as $\alpha \to 0$.

Proof sketch. For boundary point x_i at signed distance s_i from the boundary, a Gaussian probe gives, to first order, $\pi_i(\text{other side}) = \Phi(-s_i/(\sigma\alpha_{\text{eff}}))$ where α_{eff} is the effective crossing rate set by the normal component of the probe, monotone in α. Thus the map $s_i \mapsto \pi_i$ is a sigmoid of slope $\propto 1/(\sigma\,\alpha_{\text{eff}})$. The spread of $\{\pi_i\}$ across the simplex—hence $\sigma_{\min}(G)$—is maximized when this sigmoid resolves the empirical distribution of $\{s_i\}$: too small $\sigma\alpha$ saturates it (all π_i at vertices), too large flattens it (all π_i at the centroid). Setting the sigmoid scale comparable to the spread of $\{s_i\}$ yields the band; its endpoints scale inversely with α_{eff}, widening the band as α grows. □

Corollary 2 (Identifiability is a probe-relative property)**.** *Combining Proposition 1 with Theorem 2: there exist gates and instance sets that are unrecoverable under a poorly-scaled probe ($\kappa(H) = \infty$) yet recoverable under a well-scaled probe of the same family. Hence probe design changes* what is identifiable*, not only the estimation error at fixed identifiability.*

8 Anchor impurity and graceful degradation (T4)

We now let a fraction ε of anchors be impure—routed to the wrong mode—modeling a (static) adversary who corrupts the one coordinate recovery trusts most, and bridging to the adaptive setting of future work.

Assumption 2 (ε-impure anchors)**.** Each anchor is, independently, pure with probability $1 - \varepsilon$ and drawn from a wrong mode with probability ε.

Proposition 2 (Degradation and the inversion floor)**.** *Under Assumptions 1(i,ii) and 2, using anchors as the pure-instance witnesses:*

1. *(**Graceful regime**) For $\varepsilon < \varepsilon^\star$ with $\varepsilon^\star$ depending on $\sigma_{\min}(W)$ and the margin, the SPA vertex set is unchanged up to $\mathcal{O}(\varepsilon\ \mathrm{diam}(W))$ perturbation, and $\max_i \|\widehat{\pi}_i - \pi_i\| = \mathcal{O}(\epsilon\kappa^2(W) + \varepsilon)$.*

2. *(**Permutation survival**) The recovered partition agrees with the truth* up to a global relabeling *for all $\varepsilon \neq \frac{1}{2}$; i.e. orientation, not partition structure, is what degrades. As $\varepsilon \to 1$ the labels invert ($\widehat{z}_i \to \rho_{\text{flip}}(z_i)$), and permutation-corrected accuracy returns toward* 1.

3. *(**Identifiability floor**) At $\varepsilon = \frac{1}{2}$ the anchor carries no orientation information; the mode labels are unidentifiable from anchors alone (though the unlabeled partition may still be recoverable from H's geometry).*

Proof sketch. (1) A ε-fraction of mislabeled pure witnesses moves each estimated vertex $\widehat{w}_k$ by a convex combination weight ε toward other vertices; for ε below the margin that separates the vertex clusters, the max-norm selection still picks one representative per true vertex, with displacement $\mathcal{O}(\varepsilon \operatorname{diam}(W))$. Propagation as in Theorem 2 adds the ε term. (2)–(3) The anchor acts as a noisy orientation label with flip rate ε; the sign of its informativeness is $\operatorname{sign}(1 - 2\varepsilon)$, which is nonzero off $\varepsilon = \frac{1}{2}$, giving a consistent (possibly inverted) labeling, and vanishes at $\frac{1}{2}$. □

Remark 6 (Bridge to the adaptive adversary). ε is a static parameter here. The follow-up problem lets an *adaptive* adversary choose which anchors to corrupt, subject to a utility constraint (it must still deliver biased behavior on its targets). The empirical defense—true crossings are one-sided in $\operatorname{sign}(\delta)$, whereas corrupted anchors flag small-$\|\delta\|$ coordinates symmetrically—is what a memoryless adversary cannot fake, since it relies on the auditor's own probe randomness. Formalizing the resulting probe-vs-gate game is out of scope for this paper.

9 Inferring K

In the benign MoE, K is unknown. With separability, K equals the number of extreme rays of $\operatorname{cone}(M)$, estimable by the stopping rule of SPA-type algorithms: add vertices while the residual of the simplex fit exceeds the noise floor ϵ of Theorem 2. Under Assumption 1 this returns the true K once the residual drops below $\mathcal{O}(\epsilon\, \kappa^2(W))$; over-/under-estimation is controlled by the gap between the K-th and $(K{+}1)$-th selected residuals (an eigengap-style condition on W). We state recovery for known K in the main theorems and treat K-inference as this residual-threshold corollary.

10 Scope, and what each theorem decides

- **T1** (Def. 2): the generative model. New as a framing—perturbation responses as an induced separable admixture with a pure anchor and geometry-coupled weights.
- **T2** (Thm 1): the partition is identifiable; the problem is well-posed. *Decides whether the project is worth attempting.*
- **T2′** (Thm 2): a polynomial-time estimator attains it under explicit conditioning, with *no* i.i.d. assumption on the compositions. *Decides whether the method works in principle.*
- **T3** (Prop. 1): conditioning, hence recoverability, is set by probe scale σ and alignment α—probe design changes identifiability. The single-axis (α) view unifies the scaffold and MoE instantiations.
- **T4** (Prop. 2): graceful degradation under anchor impurity, with partition survival up to relabeling and an inversion floor at $\varepsilon = \frac{1}{2}$. The bridge to the adaptive-adversary sequel.

Remark 7 (Distributional generalization). Replacing the point masses (1) by genuine distributions R_k changes only the emission summary $w_k = \mathbb{E}[\psi(R_k)]$ and the within-mode variance σ_w^2; Theorems 1–2 carry through with w_k the mode means, provided the means remain separated ($\sigma_{\min}(W) > 0$). Behavioral equivalence is then exactly the collapse $w_k = w_{k'}$: modes indistinguishable in mean response under ψ are unidentifiable, the fundamental ceiling.